Lee Charles

Software Engineer • Full Stack (React | Java/Spring) • Real-Time & Data Systems
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PROFESSIONAL SUMMARY

Full Stack Software Engineer building production React and Spring Boot systems: enterprise SSO portals, real-time WebSocket architectures, and multi-database federation. Took a flagship application from SAML2/Azure AD authentication through a four-way database federation and live state sync, then distilled its infrastructure into a reusable template that measurably accelerated the next build. Owns each project end-to-end – wireframes and UI/UX design, schema, and concurrent backend processing through to shipped React UI.

CORE SKILLS

Frontend: React 19, Hooks, WebSocket/STOMP, Recharts, KendoReact, SheetJS (XLSX), react-joyride, custom design-token CSS systems
Backend: Java 21, Spring Boot (REST, JPA, Security, SAML2, WebSocket), Async/Concurrent (CompletableFuture, ExecutorService, BlockingQueue), JUnit
Data: PostgreSQL, Oracle, four-way multi-datasource federation, complex native SQL/CTEs
Design: Wireframing, rapid POC prototyping, UI/UX design systems, dark/light theming
DevOps & Other: Docker, Nginx, Jasypt, Swagger/OpenAPI, Apache PDFBox, SMB2/3 (smbj)

EXPERIENCE

First Electric Cooperative Corp (FECC) – Software Developer, Analytics & Data Engineering *2025 – Present*

- Served as lead UI/UX designer on every project below – wireframe, POCs, and production UI – without a dedicated design resource; hand-built **First2Portal's** entire component library and design-token theming system from scratch.
- Built **First2Portal**, an enterprise-wide application hub with SAML2/Azure AD SSO, federating four databases (Postgres × 2, Oracle × 2) behind ~94 REST endpoints across 16 controllers, and pushing live state to clients over WebSocket/STOMP with concurrency-safe mutation guards.
- Designed the platform's gamification engine (dense-ranked leaderboards, dedup windowing, self-healing scheduled jobs) and extracted its auth, datasource, and UI-shell infrastructure into **First2Web**, a reusable template that cut roughly two sprints of setup work off the next production build.
- Built **First2OnCall's** shift-splitting scheduling engine (date-range cuts into one to three records with exact timestamp preservation) and a crash-safe PDF backup pipeline (SMB2/3, atomic rename-promote-write rotation) for the Dispatch Team.
- Created **GridDefinition**, a from-scratch React grid engine replacing Kendo UI and now the shared reporting foundation for AMI Hourly, OMS, and DQ dashboards; engineered the AMI Hourly dashboard's Oracle/Postgres federation and statistical conversion-ratio analyzer.
- Shipped the **Daily Traffic Light** monitor (80+ reports via lightweight COUNT(*) endpoints), OMS's async .std model-ingestion pipeline, and a multi-threaded MIS PDF -> TIFF pipeline that cut manual prep from hours to minutes.

ADDITIONAL EXPERIENCE

UAMS – IT Associate / Software Tool Developer (Python GUI + VBA) *2023 – 2024*

- Built a Python GUI and automated indexer to query 7,800+ paths / 65,000+ contacts; reduced query time from ~5 minutes to under 1 second.

Pittsburgh Doors & Closets – Frontend Web Developer (Internship) *2024*

- React UI overhaul, bug fixes, SEO, and documentation for a production website.

EDUCATION

B.S., Information Technology (Minor: Software Development), Capella University – 3.85 GPA, Summa Cum Laude